

Lecture

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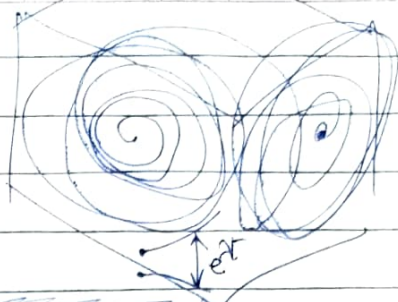
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* Lorenz equations:

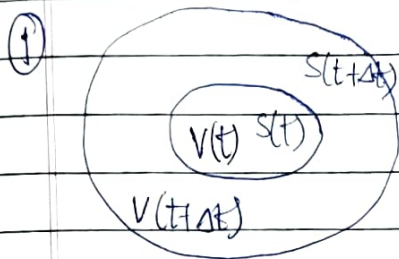
$$\dot{x} = \sigma(y-x)$$

$$\dot{y} = rx - y - xz$$

$$\dot{z} = xy - bz$$



* Observation: For $\sigma, r, b > 0$, $(x, y, z) \rightarrow (-x, -y, z)$ this is a symmetry



$$V(t+dt) = V(t) + \int_S (\mathbf{f} \cdot \hat{n}) dt \cdot dA$$

$$dV = \int_V (\nabla \cdot \mathbf{f}) dV$$

$$\Rightarrow \frac{dV}{dt} = \int_V (\nabla \cdot \mathbf{f}) dV$$

$$= \frac{\partial}{\partial x} (\sigma(y-x))$$

$$+ \frac{\partial}{\partial y} (rx - y - xz)$$

$$+ \frac{\partial}{\partial z} (xy - bz)$$

$$\Rightarrow \frac{dV}{dt} = -\sigma - 1 - b < 0 \quad \forall \sigma, r, b > 0$$

So, system is dissipative

② Fixed points:-

$$\bar{y} = \bar{x}$$

$$r\bar{x} - \bar{x} - \bar{x}z = 0 \Rightarrow \boxed{\bar{x} = 0} \text{ or } \boxed{\bar{z} = r-1}$$

$$\bar{x}\bar{y} - b\bar{z} = 0 \Rightarrow \bar{x}^2 = b(r-1) \Rightarrow \boxed{\bar{x} = \pm \sqrt{b(r-1)}}$$

$$\Rightarrow \text{f.p. } (0, 0, r-1); (\sqrt{b(r-1)}, \sqrt{b(r-1)}, r-1); (-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r-1)$$

exist if $r-1 > 0 \Rightarrow r > 1$

③ Linear Stability:—

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} = \begin{pmatrix} -\sigma + \sigma & 0 \\ r-2 & -1 & -\alpha \\ y & \alpha & -b \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$(\bar{x}, \bar{y}, \bar{z})$

$$\Rightarrow J(0,0,0) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{pmatrix}$$

z-direction is completely decoupled

$$A_{xy} = \begin{pmatrix} -\sigma & \sigma \\ r & -1 \end{pmatrix}$$

$$\Rightarrow z(t) = e^{-bt} z(0)$$

always attractive

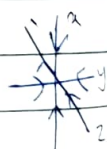
$$\Rightarrow \det(A_{xy}) = \sigma(1-r)$$

$$\text{Tr}(A_{xy}) = -(\sigma + 1) < 0$$

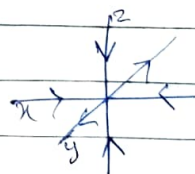
→ For $r < 1$, $\det > 0$.

$$\begin{aligned} \text{Tr}^2 - 4\det &= (\sigma^2 + 2\sigma + 1) - 4\sigma(1-r) \\ &= \sigma^2 + 2\sigma + 1 - 4\sigma + 4\sigma r \\ &= (\sigma - 1)^2 + 4\sigma r > 0 \Rightarrow \text{sink} \end{aligned}$$

(in all directions)



→ For $r > 1$, $\det < 0 \rightarrow$ saddle



→ For $r = 1$, $\det = 0$ Attract

etc exist in this region only

$\{0\}$

$r=1$

④ Global Stability:— (For $(0,0,0)$)

$$V(x,y,z) = \frac{1}{\sigma} x^2 + y^2 + z^2$$

→ $V(x,y,z) > 0$ everywhere except at origin

~~$$\therefore \frac{dV}{dt} = \frac{2x}{\sigma} + 2y + 2z$$~~

$$\Rightarrow \frac{dV}{dt} = \frac{2x}{\sigma} \sigma(y-x) + 2y(rx-y-3x) + 2z(xy-bz)$$

$$= -\left[x - \frac{\sigma+1}{2}y\right]^2 - \left[1 - \left(\frac{\sigma+1}{2}\right)^2\right]y^2 - bz^2 < 0$$

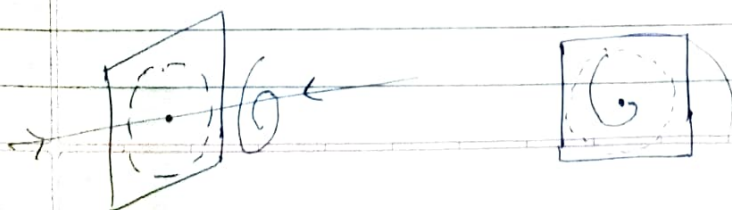
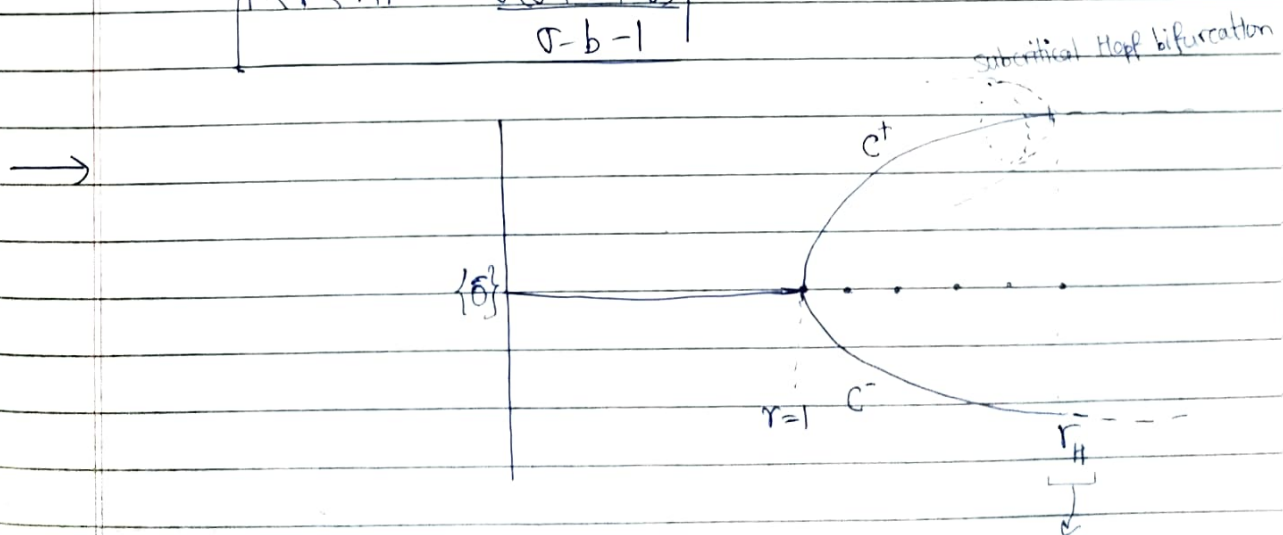
⇒ Globally stable

⑤ Other two fps—

- For the other two fps (c^+ & c^-),
 $\{\lambda\} = \{\text{real}, \pm i\omega\}$

⊗ Check for what value of r eigenvalues become $\pm i\omega$

$$1 < r < r_H = \frac{\sigma(\sigma+b+3)}{\sigma-b-1}$$



this is called a saddle cycle

⑥ Trapping Region:—

$$T := W(x, y, z) = rx^2 + \sigma y^2 + \sigma(z-2r)^2 \leq c$$

$$\Rightarrow \frac{dW}{dt} = 2\sigma[-rx^2 - y^2 - bz^2 + 2rbz]$$

→ We want to check where it is < 0

$$\Rightarrow \frac{dW}{dt} < 0 \Rightarrow rx^2 + y^2 + bz^2 - 2rbz > 0$$

$$\Rightarrow \boxed{rx^2 + y^2 + b(z-r)^2 > br^2}$$

ellipsoid ← trapping region

ellipsoid:

$$\textcircled{E} E := rx^2 + y^2 + b(z-r)^2 > br^2$$



Take c finite and large enough so that T engulfs E .

→ Now we had

$$T := rx^2 + \sigma y^2 + \sigma(z-2r)^2 \leq c$$

$$\Rightarrow \boxed{T := \frac{x^2}{\frac{c}{r}} + \frac{y^2}{\frac{c}{\sigma}} + \frac{(z-2r)^2}{\frac{c}{\sigma}} = 1}$$

$$\boxed{E := \frac{x^2}{br} + \frac{y^2}{br^2} + \frac{(z-r)^2}{r^2} = 1}$$

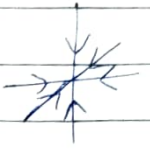
At $(y=0, z=r)$, $\bar{x}_E = \pm\sqrt{br}$
 $\bar{x}_T > \bar{x}_E \Rightarrow \underline{cx \geq r^2(\sigma+b)}$

At $(x=0, z=r)$; $\bar{y}_E = \pm\sqrt{br^2}$
 Again, put in T , get
 $\bar{y}_T > \bar{y}_E \Rightarrow \underline{cy \geq \sigma r^2(b+1)}$

At $(x=0, y=0)$; $\bar{z}_E = 0, 2r$,
 Again, put in T , get
 $\underline{cz \geq 4\sigma r^2}$

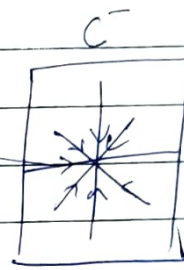
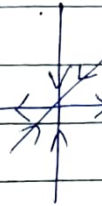
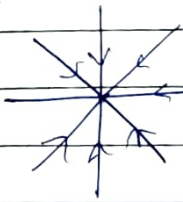
So, our c should be set as
 $c = \max\{c_x, c_y, c_z\}$

⑦. At $r < 1$:- Only fp is $(0,0,0)$, is globally stable

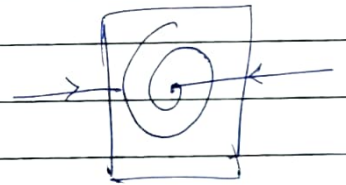
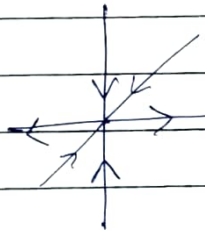
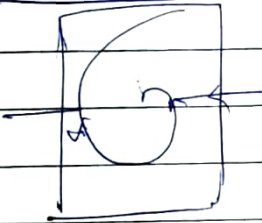


• At $r > 1$:-

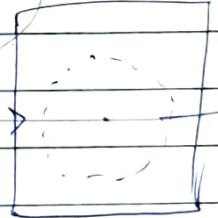
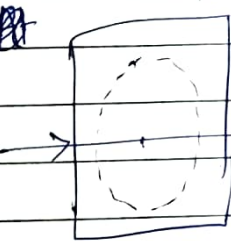
c^+



• At $r < r_H$:-



• ~~At~~



Chaotic behavior

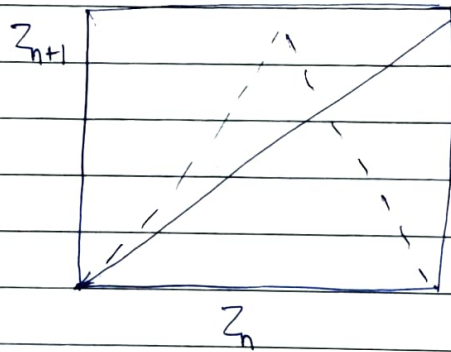
* Chaos & Chaotic behavior:-

- Aperiodic long-time behavior of a deterministic system
- Trajectories don't converge $\rightarrow$ xfp. ~~(periodic/quasiperiodic)~~
 $\times$ periodic
 $\times$ quasiperiodic

- Sensitive dependence in initial condition
 $\Rightarrow$ exp divergence

- Trajectories are bounded $\subset T$
- Butterfly-like objects observed (attractor) - strange object
- Dimension ≥ 3 required for this to be observed

* Function z_n :-



$$z_{n+1} = f(z_n)$$